THE UNIVERSITY OF UTAH®

(ECE 6545)

Project Report

MRI Image Reconstruction Project

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1 Introduction and Motivation

Late Gadolinium Enhancement (LGE) MRI has become an important technique for visualizing and quantifying tissue characteristics in cardiac imaging [5]. This modality is particularly valuable for its ability to highlight regions of myocardial scarring, fibrosis, and other tissue abnormalities by exploiting the differential uptake and washout of gadolinium-based contrast agents in normal versus damaged tissue. On the other hand, myocardial perfusion assessment is a crucial enhancement to normal clinical protocols, essential for evaluating not just vascular diseases but other conditions that may indirectly affect circulatory elements, such as microvascular function [3]. MRI can detect decreased vascular supply following myocardial infarction, as well as quantitatively map capillary myocardial blood flow (MBF) during rest and stress. The first-pass dynamic contrast-enhanced (DCE) MRI is currently accessible on scanners. Perfusion imaging is basically T1-weighted images after a contrast agent Gadolinium (Gd) passes.

MRI reconstruction from undersampled k-space data represents a significant challenge in the field of medical imaging. Traditional LGE-MRI acquisitions face fundamental limitations:

- Extended acquisition times that limit throughput and accessibility
- Susceptibility to motion artifacts during lengthy scans
- Compromises between spatial resolution, signal-to-noise ratio, and scan duration

Whereas, conventional perfusion MRI acquisitions and reconstruction face challenges:

- Cardiac motion artifacts when it is not ECG gated
- Respiratory motion artifacts when the patients don't hold or shallow their breath
- Not robust against different protocols and acquisition schemes
- Dependency on temporal feature heavily affects the reconstruction algorithm

These constraints have motivated extensive research into accelerated imaging techniques. Conventional approaches such as parallel imaging and compressed sensing have made significant advances, but they often rely on hand-crafted priors that may not fully capture the complex structure of medical images. As acceleration factors increase beyond 2-3$\times$, these methods frequently exhibit diminishing returns in reconstruction quality. Recent advances in deep learning have shown promise for MRI reconstruction, but most approaches depend on supervised learning with large datasets of paired undersampled and fully-sampled images. This requirement presents several practical challenges:

- Difficulty in obtaining high-quality, fully-sampled reference images
- Limited generalizability across different acquisition protocols and anatomical variations
- Potential for learning and propagating systematic artifacts present in training data

Deep Image Prior (DIP) offers an intriguing alternative that addresses many of these limitations. Rather than learning from a large dataset, DIP leverages the structure of convolutional neural networks themselves as an implicit regularizer. This approach has several advantages for MRI reconstruction:

1. It requires no training data, operating in an unsupervised manner
2. It adapts specifically to each input image, potentially capturing unique anatomical features
3. It can be integrated with the physics of MRI acquisition through data consistency constraints

Despite these advantages, vanilla DIP suffers from stability issues and tendency to overfit to noise in later iterations. Self-Guided DIP addresses these limitations by incorporating a self-regularization mechanism that promotes consistency under random perturbations of the network input. This work explores the application of Self-Guided DIP for reconstructing high-quality MRI from significantly undersampled acquisitions. Our goal is to develop a reconstruction framework that achieves substantial acceleration factors (4-6×) while preserving the anatomical details necessary for accurate image analysis. Through this approach, we aim to contribute to the broader goal of making advanced MRI techniques more accessible and efficient without compromising diagnostic quality.

2 Method

2.1 Self-Guided Deep Image Prior Framework

The Self-Guided Deep Image Prior approach represents a significant advancement over traditional DIP methods for MRI reconstruction [1]. At its core, this method leverages both the implicit regularization provided by the neural network architecture and a self-guided regularization mechanism. The fundamental concept behind DIP is to parameterize the reconstructed image through an untrained convolutional neural network. Rather than learning from a dataset of example images, the network parameters are optimized for each specific undersampled input [4]. The basic optimization problem for MRI reconstruction using DIP can be expressed as:

$$\arg\min_{\theta} \sum_{c=1}^{N_c} |A_c f_\theta(z) - y_c|_2^2 \tag{1}$$

where f_θ represents the neural network with parameters θ, z is the network input (typically random noise), A_c is the forward model incorporating the undersampling mask, coil sensitivities, and Fourier transform for coil c, and y_c is the acquired k-space data for coil c. The Self-Guided DIP extends this framework by incorporating a self-regularization mechanism. The optimization problem becomes:

$$\arg\min_{\theta,z} \sum_{c=1}^{N_c} |A_c E_\eta[f_\theta(z+\eta)] - y_c|2^2 + \alpha|E_\eta[f_\theta(z+\eta)] - z|_2^2 \tag{2}$$

where η represents random perturbations and E_η denotes expectation over these perturbations. The second term serves as a self-guidance mechanism, encouraging the network output to be consistent under input perturbations. This approach addresses two key limitations of vanilla DIP:

1. Overfitting to noise in later iterations
2. Instability during the optimization process

The final reconstruction is obtained through:

$$x^* = E_{\eta\sim P_\eta}[f_\theta(z+\eta)] \tag{3}$$

where η is sampled from distribution P_η (typically Gaussian).

2.2 Network Architecture

For implementing Self-Guided DIP, we employ a ResUNet architecture with skip connections, which has demonstrated strong performance in medical image processing tasks [2]. The network consists of:

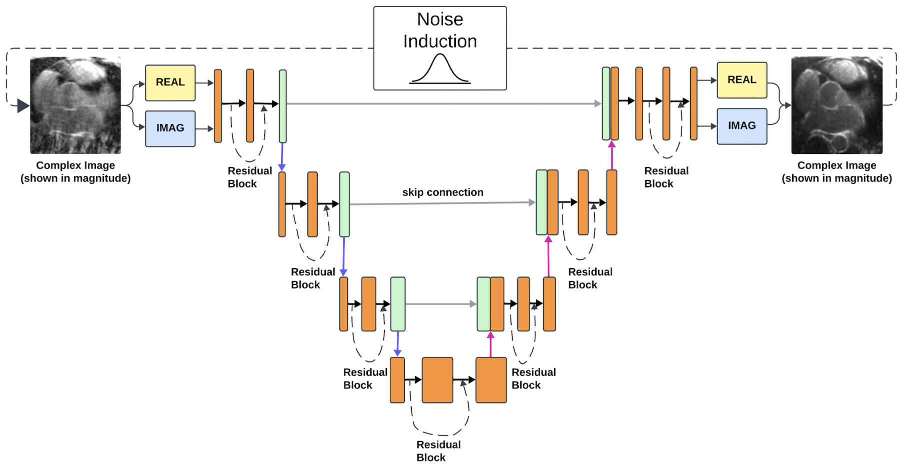


Figure 1: Network Architecture

- An encoder path with convolutional layers and downsampling operations
- A decoder path with upsampling operations and convolutional layers
- Skip connections between corresponding encoder and decoder layers
- Residual blocks containing two convolutional layers with batch normalization

The network processes complex-valued input data by treating real and imaginary components as separate channels.

2.3 Optimization Strategy

The optimization procedure for Self-Guided DIP requires careful consideration to achieve optimal results:

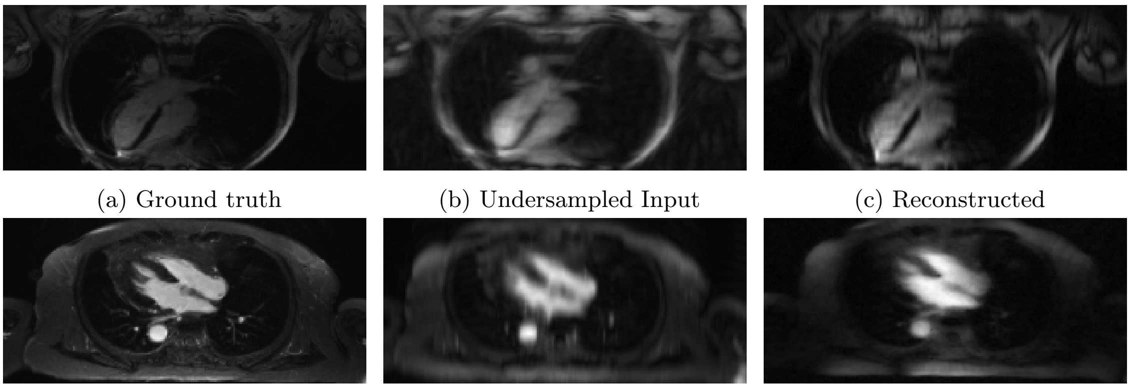
1. The network is initialized with random weights.
2. The input z can be either random noise or a zero-filled reconstruction.
3. During each iteration, we add Gaussian noise perturbations η to the input.
4. The Adam optimizer with an appropriate learning rate (typically 3×10^{-4}) is used to update the network parameters.
5. A key consideration is determining appropriate stopping criteria, as DIP models tend to overfit to noise when trained for excessive epochs.
6. The regularization weight α balances the data consistency term with the self-guidance term.

This approach allows for patient-specific optimization without requiring training on large datasets. By leveraging the network architecture as an implicit regularizer and incorporating self-guidance, the method can effectively handle highly undersampled data while preserving important anatomical details critical for diagnostic assessment.

3 Dataset

The dataset used for the work is released for a competition, which is named CMRxRecon25 [6]. The CMRxRecon2025 (Towards Foundation Model) challenge is a part of the 28th MICCAI 2025. The CMRxRecon25 dataset is a unique cardiac MRI benchmarking dataset with real-world clinical scenarios. It has dataset from multi-center, multi-vendor, and multiple diseases. Current vital progression of AI in accelerated MRI enabling in specific contrast type or sampling pattern. Because of the aforementioned diversities, distribution mismatches between the training and target data are frequently unavoidable. Consequently, one of the most important technological problems for multi-parametric CMR imaging is the development and validation of universal and reliable reconstruction models for managing these diversities. The CMRxRecon has diversity due to 12 MRI modalities (we are working with only two—LGE and perfusion), five different centers in different locations, 10+ different scanners, healthy and unhealthy (five diseases) population, and three types of sampling patterns (gaussian, uniform, radial) with three (8x, 16x, and 24x) types of acceleration factors. Unlike previous versions of CMRxRecon, this time there were 600+ subjects while collecting the dataset.

4 Results



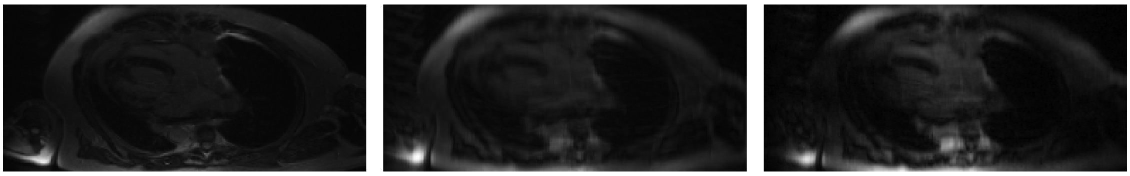
(a) Ground truth (b) Undersampled Input (c) Reconstructed

(d) Ground truth (e) Undersampled Input (f) Reconstructed

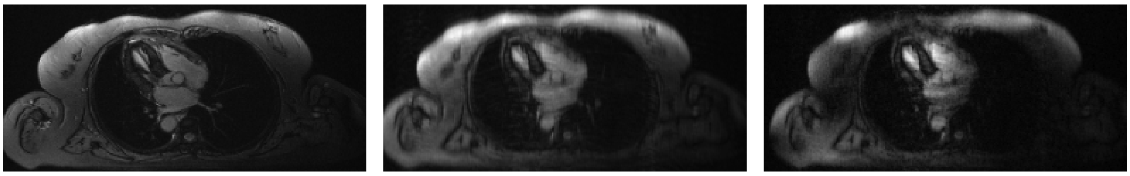
Figure 2: Performance of Self-guided DIP on different samples from the CMRxRecon25 LGE dataset. Two different slices represent two different scanners (Top Row: Siemens and Bottom Row: UIH) examples. We have used cartesian 24X undersampling for this one. Both the data are from the same center.

We explored the robustness of self-guided DIP through reconstructing MRI raw k-space data with the variability of centers and scanners. Different centers may impose different acquisition protocol, and different scanners have different hardwards. Thus, we experimented self-guided DIP's ubiquity with the LGE data from different centers and scanners. From Figure 2, we found that the reconstructed images from the Siemenes scanner showed good imporvements over the noisy input. Whereas, the reconstructed images from UIH scanner also showed significant improvement over undersampled input, despite being blurrier than the ground truth. For different centers data from Figure 3, we see that the reconstructions are less pronounced compared to the ground truth, but much sharper than the undersampled input; thus potentially proving the robustness of self-guided DIP in these applications.

The universality of self-guided DIP can also be shown from the robustness along different sampling patterns. Because most of the traditional methods depend on sampling patterns, but from Figure 4, similar results are seen without degradation. The undersampled mask multiplied by the fully sampled k-space (ground truth) is the input of the ResUnet. Also, from the table 1, in perfusion, evaluation metrics show comparable results like the qualitative consistency.

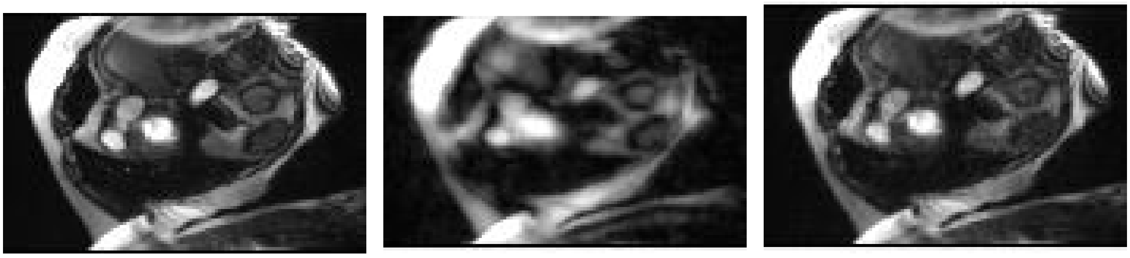


(a) Ground truth (b) Undersampled Input (c) Reconstructed



(d) Ground truth (e) Undersampled Input (f) Reconstructed

Figure 3: Performance of Self-guided DIP on different samples from the CMRxRecon25 LGE dataset. Two different slices represent two different centers (anonymized) examples. We have used cartesian 24X undersampling for this one. Both the data are from the same scanner.



(a) Ground truth (b) Undersampled Input (c) Reconstructed

(d) Ground truth (e) Undersampled Input (f) Reconstructed

Figure 4: Performance of Self-guided DIP on different sampling patterns on the CMRxRecon25 perfusion dataset. All frames are from same patient and same time frame. Undersampling acceleration factor of masks were 24x. The top row is for radial sampling mask and the bottom row for cartesian (uniform) sampling mask.

Table 1: Quantitative performance on different cases

Robustness factor	Different set	PSNR	SSIM	NRMSE
Different sampling masks (Perfusion)	Cartesian	33.46	0.93	0.02
	Radial	34.35	0.91	0.01
Different Scanners (LGE)	Center 001	31.28	0.89	0.07
	Center 002	30.75	0.87	0.04
Different Centers (LGE)	Siemens	29.64	0.85	0.05
	UIH	28.92	0.83	0.06

5 Discussion

The Self-Guided Deep Image Prior demonstrates encouraging robustness across diverse MRI scenarios. Our approach successfully handles variations in sampling patterns for perfusion imaging, with both Cartesian and Radial trajectories yielding good reconstruction quality. Despite some performance decline across different centers and scanners for LGE images, the method maintains reasonable image quality, preserving essential diagnostic information. This robustness to acquisition diversity is promising for clinical applications where scanner and protocol variations are common.

However, to fully optimize performance, particularly for LGE images, additional hyperparameter tuning and regularization weight adjustment are necessary. With these refinements, the approach has potential to provide consistent reconstruction across various clinical settings. Future work should address these challenges through improved regularization strategies and more systematic approaches to parameter selection that can adapt to variations in acquisition characteristics.

References

[1] Evan Bell et al. "Robust self-guided deep image prior". In: *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2023, pp. 1–5.

[2] Foivos I Diakogiannis et al. "ResUNet-a: A deep learning framework for semantic segmentation of remotely sensed data". In: *ISPRS Journal of Photogrammetry and Remote Sensing* 162 (2020), pp. 94–114.

[3] Geon-Ho Jahng et al. "Perfusion magnetic resonance imaging: a comprehensive update on principles and techniques". In: *Korean journal of radiology* 15.5 (2014), pp. 554–577.

[4] Shijun Liang et al. "Analysis of deep image prior and exploiting self-guidance for image reconstruction". In: *IEEE Transactions on Computational Imaging* (2025).

[5] Christopher J McGann et al. "New magnetic resonance imaging-based method for defining the extent of left atrial wall injury after the ablation of atrial fibrillation". In: *Journal of the American College of Cardiology* 52.15 (2008), pp. 1263–1271.

[6] Chengyan Wang et al. "CMRxRecon: A publicly available k-space dataset and benchmark to advance deep learning for cardiac MRI". In: *Scientific Data* 11.1 (2024), p. 687.